Lecture 14: 4-vectors continued

Based on Morin's *Introduction to Classical Mechanics*, Ch. 13.4 –13.6

Outline

1. Recap lecture 13
2. Energy and Momentum
3. Force and Acceleration
4. Physical Laws

1 Recap lecture 13

In Lecture 13, we defined 4-vectors. We explained their properties and provided examples. In SR, the most important 4-vectors are the spacetime interval $dS = (dt, dx, dy, dz)$, the 4-velocity $V = (\gamma, \gamma\mathbf{v})$ and the 4-momentum $P \equiv mV = (m\gamma, m\gamma\mathbf{v}) = (E, \mathbf{p})$. By starting with the spacetime interval and using successive derivatives on the spacetime interval dS with respect to proper time $d\tau$, we can construct an infinite tower of 4-vectors. Taking derivatives with respect to τ or multiplying by m (as is done for the 4-momentum) results in a new 4-vector, as both $d\tau$ and m are frame-independent quantities (and hence should not be affected by a Lorentz transformation).

Using 4-vectors, you find that there is a natural definition of relativistic energy and (3) momentum, as the time and spatial components of the 4-momentum vector. While we used heuristic arguments to build these quantities in Lecture 11, we obtained them here by simple manipulation of the spacetime interval 4-vector.

Similarly, by taking yet another derivative, either from V or P, with respect to τ, we found that you can find an expression for the 4-acceleration and the 4-force, respectively. The spatial components of the 4-force are equivalent to the force we derived in lecture 12, showing consistent results. Furthermore, the 4-force F can also be related to the 4-acceleration A as $F = mA$, which again shows consistency with our idea of Newton's notion of forces.

We also introduced a bit of tensor notation, e.g.

$$dS^2 = \eta_{\mu\nu} dx^\mu dx^\nu. \tag{1}$$

Combined with the tensor notation for Lorentz transformations, this allowed us to find some formal proofs of the nature and properties of 4-vectors. We do not expect you to fully comprehend the tensor algebra presented in this course. It serves as a complementary component to the details shown in Morin's book.

2 Energy and Momentum

Since P is a 4-vector and we argued the norm of a 4-vector is frame-independent, you could move to a frame where an object is at rest ($P(v \neq 0) \rightarrow P'(v' = 0)$). In this frame, we have $P' = (m, 0, 0, 0)$, and hence $P'^2 = m^2 = P^2$. We already knew this, of course, but it is nice to see how this could have been derived very easily by a simple transformation.

If you have a collection of particles, knowledge of the norm is useful. Suppose you have a collection of particles n, then for any subset of particles $m < n$, we know

$$\left(\sum_{i=1}^{m} E\right)^2 - \left(\sum_{i=1}^{m} \mathbf{p}\right)^2 = \text{invariant.} \tag{2}$$

This is true because it presents the squared norm of the sum of the energy-momentum 4-vectors of the subset m of n particles. As we had shown in lecture 13, a linear combination of 4-vectors is again a 4-vector.

In the center of mass (CM) frame (lecture 9) $\sum \mathbf{p} = 0$, the above would reduce to the square of the energies, which reduces to m^2 for a single particle.

Since P is a 4-vector, according to the definition of a 4-vector, its components should transform according to the Lorentz transformations, i.e., (motion β in the x direction)

$$\begin{aligned} E &= \gamma(E' + \beta p'_x), \\ p_x &= \gamma(p'_x + \beta E'), \\ p_y &= p'_y, \\ p_z &= p'_z. \end{aligned} \tag{3}$$

This is equivalent to the transformation rules we derived in lectures 10 and 11.

Since E and $\mathbf{p}$ are part of the same 4-vector, if one of the two is conserved in every frame in a collision, then so is the other. To see this, you can consider the 4-momentum change 4-vector $\Delta P \equiv P_{\text{after}} - P_{\text{before}}$. If, e.g., E is conserved in every frame, then $\Delta P_0 = 0$ in every frame. By virtue of one of the theorems discussed in the previous lecture, we know that the other component should be zero, and hence $\mathbf{p}$ should also be conserved in every frame.

3 Force and Acceleration

We will assume m is constant and does not depend on time. Recall the 4-force in an instantaneous inertial frame S' where $v' = 0$:

$$F' = \gamma\left(\frac{dE'}{dt}, \mathbf{F}'\right) = (0, \mathbf{F}'). \tag{4}$$

Next, consider another frame S, in which the particle moves with velocity v ($= \beta$) in the x direction. Since F' is a 4-vector, it transforms according to the Lorentz transformations:

$$\begin{aligned} F_0 &= \gamma(F'_0 + \beta F'_1) = \gamma\beta F'_x, \\ F_1 &= \gamma(F'_1 + \beta F'_0) = \gamma F'_x, \\ F_2 &= F'_2 = F'_y, \\ F_3 &= F'_3 = F'_z. \end{aligned} \tag{5}$$

In addition, we also know that

$$\begin{aligned}
F_0 &= \gamma \frac{dE}{dt}, \\
F_1 &= \gamma F_x, \\
F_2 &= \gamma F_y, \\
F_3 &= \gamma F_z.
\end{aligned} \tag{6}$$

We can now equate the two, to find

$$\begin{aligned}
\frac{dE}{dt} &= \beta F'_x, \\
F_x &= F'_x, \\
F_y &= F'_y/\gamma, \\
F_z &= F'_z/\gamma.
\end{aligned} \tag{7}$$

This is the same as we found in lecture 12, when we showed that the force in the longitudinal direction (x) remains the same, while the force along the transverse direction (y, z) is amplified by a factor γ in the particle frame S'.

The ratio of $F_y/F_x = \gamma^{-1}F'_y/F'_x$, i.e. the ratio of the forces decreases by a factor γ when going from S' to S (hence the norm of the vector (F_x, F_y) will decrease).

The first line combined with the second line tells us that $dE = F_x dx$, which is the work-energy result we derived in lecture 12 and is analogous to the result for classical motion in lecture 9.

Note that we can not write $F'_y = F_y/\gamma$ (by switching the two frames). This is because when considering forces on the particle, there is one specific frame, the particle frame (S'), where the forces apply, which is part of the underlying assumption when the 4-vectors were derived.

We can apply the same analysis for the acceleration 4-vector. In frame S', the instantaneous inertial frame of the particle, we have

$$A' = (0, \mathbf{a}'), \tag{8}$$

because $v' = 0$ in S'. In the frame S, where the particle is moving with velocity $v = \beta$ in the x direction, we can relate the components of A to those of A' using the Lorentz transformations:

$$\begin{aligned}
A_0 &= \gamma(A'_0 + \beta A'_1) = \gamma\beta a'_x, \\
A_1 &= \gamma(A'_1 + \beta A'_0) = \gamma a'_x, \\
A_2 &= A'_2 = a'_y, \\
A_3 &= A'_3 = a'_z.
\end{aligned} \tag{9}$$

From the expression for the acceleration 4-vector derived in lecture 13, we also have

$$\begin{aligned}
A_0 &= \gamma^4 v a_x, \\
A_1 &= \gamma^4 a_x, \\
A_2 &= \gamma^2 a_y, \\
A_3 &= \gamma^2 a_z.
\end{aligned} \tag{10}$$

Equating the two again, yields:

$$\begin{aligned} a_x &= a'_x/\gamma^3, \\ a_y &= a'_y/\gamma^2, \\ a_z &= a'_z/\gamma^2. \end{aligned} \tag{11}$$

We recover the same results found in Lecture 12. In contrast to the force transformation, we find $a_y/a_x = \gamma a'_y/a'_x$. This means that $\mathbf{F} = m\mathbf{a}$ does not generally apply in SR. If it were true in one frame, it would not be true in another.

Example 1: Acceleration for circular motion

Consider a particle moving around a circle $x^2 + y^2 = r^2$ $(z = 0)$ with constant speed as observed in a lab frame S.[1] At the instant the particle crosses the negative y axis, find the three-acceleration and four-acceleration in S and in the instantaneous inertial frame comoving with the particle.

First, we write for the radial position of the particle at some instant

$$\mathbf{r} = r(\hat{x}\cos\omega t + \hat{y}\sin\omega t), \tag{12}$$

with ω the angular velocity. To get the velocity, we differentiate with respect to t:

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} = \omega r(-\hat{x}\sin\omega t + \hat{y}\cos\omega t). \tag{13}$$

Hence we have $|\mathbf{v}| = v = \omega r$. To obtain the acceleration, we differentiate again:

$$\mathbf{a} = \frac{d\mathbf{v}}{dt} = -\omega^2 r(\hat{x}\cos\omega t + \hat{y}\sin\omega t). \tag{14}$$

Thus we find $|\mathbf{a}| = a = v^2/r$. In the moment that the particle is crossing the negative y axis, the acceleration is zero in the x direction, and all the acceleration is in the positive y direction. With $\Theta = \omega t$, we have $\Theta = \frac{3\pi}{2}$ and hence

$$\mathbf{a} = (0, v^2/r, 0). \tag{15}$$

From Eq. (10), we then find

$$A = (0, 0, \gamma^2 v^2/r, 0). \tag{16}$$

To obtain the coordinates in the S' frame, we use that the components of A are related to the components of A' through a Lorentz transformation in the x direction. This means that the $A_2 = A'_2$ (and all the other components that are zero remain zero). Thus, we have:

$$A' = (0, 0, \gamma^2 v^2/r, 0) \tag{17}$$

The 3 acceleration in S', where $v' = 0$ and $\gamma' = 1$, should be related to the spatial components of A':

$$\mathbf{a}' = (0, \gamma^2 v^2/r, 0). \tag{18}$$

This result is consistent with Eq. (11).

[1] The local relation $d\tau = \sqrt{1 - v^2}\, dt$ is valid for any timelike worldline. Here the speed v is constant, although the velocity vector $\mathbf{v}$ changes direction.

4 The form of physical laws

One of the key postulates of SR is that all inertial frames are equivalent. If a physical law holds in one inertial frame, it will hold in all frames. If not, we would be able to differentiate between frames. For example, we found that $\mathbf{F} = m\mathbf{a}$ can not be a physical law (although we long defended it to be!) since $\mathbf{F}$ transforms differently from $\mathbf{a}$ as we showed in the previous section. It might be true in one frame, but it certainly can not be true in all frames.

As a guiding principle, for a statement or physical law to hold in all frames, it should involve 4-vectors (as these transform according to the Lorentz transformations). Suppose in one frame we have a statement that relates two 4-vectors, $A = B$. Does this hold in all inertial frames? Yes, since we can apply the Lorentz transformation (and an infinite series if we absolutely want to) and find

$$A = B \rightarrow \Lambda A = \Lambda B \rightarrow A' = B', \tag{19}$$

where we have again used Λ as our Lorentz transformation matrix. This relation therefore also holds in a new frame S'. Only relations that are physically and mathematically valid in one frame are valid in all frames. For example, $P = mV$ and the four-vector equation $F = mA$ are covariant relations, whereas the three-vector equation $\mathbf{F} = m\mathbf{a}$ is not generally valid in SR.

We have also seen that scalars are frame independent, so physical laws may contain scalars such as m, or may be scalar equations such as $P \cdot P = m^2$. If they are true in one frame, they hold in all frames. In SR, we concern ourselves with scalars and four-vectors. Physical laws may also involve higher-rank tensors: scalars are rank-zero tensors, while four-vectors are rank-one tensors. Such higher-rank tensors can be built from four-vectors.

There is nothing really new here. In Newtonian mechanics, we have $\mathbf{F} = m\mathbf{a}$ as a possible physical law because both sides are 3-vectors. $\mathbf{F} = m(2a_x, a_y, a_z)$ is not a possible law because $(2a_x, a_y, a_z)$ is not a 3-vector, since it depends on which axis you label the x axis. It makes no sense for a physical law to depend on your arbitrary choice of coordinates.